

RALEIGH-DURHAM, NC, USA

WMO: 723060

Lat: 35.892N

Lon: 78.782W

Elev: 416

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13722

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.7	23.6	0.4	5.7	25.7	5.3	7.3	30.6	23.3	59.8	20.0	58.2	4.3	30	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	95.0	75.6	92.6	75.1	90.3	74.5	78.4	89.8	77.4	88.2	76.4	86.2	8.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	134.9	82.8	74.4	130.5	81.6	73.5	126.5	80.5	42.1	90.0	41.0	88.2	40.0	86.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	16.5	14.4	DB	12.3	99.0	5.3	3.0	8.5	101.1	5.4	102.9	2.4	104.6	-1.4	106.8
			WB	10.4	80.6	4.9	1.3	6.9	81.6	4.0	82.3	1.3	83.1	-2.2	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	41.4	44.7	51.5	60.8	68.7	76.5	79.9	78.5	72.3	61.5	51.2	44.7	(6)
	DBStd	15.63	10.50	9.67	9.53	8.32	7.02	5.37	4.15	4.54	5.96	7.97	8.54	9.38	(7)
	HDD50	940	308	202	100	11	0	0	0	0	0	8	88	223	(8)
	HDD65	3188	733	570	430	178	47	3	0	0	13	162	420	632	(9)
	CDD50	4970	41	54	145	334	579	796	925	882	668	366	122	58	(10)
	CDD65	1745	1	2	10	51	162	349	460	417	231	55	5	2	(11)
	CDH74	15838	2	11	92	474	1398	3285	4531	3881	1712	413	33	6	(12)
	CDH80	6413	0	0	9	104	443	1425	2088	1665	585	92	2	0	(13)
(14) Wind	WSAvg	6.0	6.3	6.6	7.2	7.6	6.6	5.9	5.7	5.2	5.5	5.1	5.2	5.5	(14)
(15) Precipitation	PrecAvg	44.1	3.4	3.3	4.0	3.1	3.8	3.7	4.7	4.3	4.2	3.1	3.1	3.3	(15)
	PrecMax	60.2	7.1	7.2	7.8	8.7	7.7	10.4	10.6	12.2	21.8	9.4	9.0	6.7	(16)
	PrecMin	32.1	0.9	0.4	1.0	0.2	0.6	0.3	0.9	0.9	0.2	0.0	0.5	0.4	(17)
	PrecStd	6.7	1.6	1.5	1.5	1.7	1.5	2.4	2.4	2.3	3.5	2.0	2.1	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.6	75.4	81.2	87.0	91.2	96.9	99.1	97.8	93.2	86.9	78.2	73.7	(19)
		MCWB	61.4	62.2	63.1	67.5	72.7	75.4	76.5	76.6	71.5	71.4	64.8	63.8	(20)
	2%	DB	67.2	70.0	76.4	83.1	87.9	93.3	95.2	93.9	89.3	82.4	73.7	69.3	(21)
		MCWB	59.7	58.5	61.2	65.5	71.1	74.7	76.3	75.8	72.4	69.3	62.2	62.5	(22)
	5%	DB	63.2	65.8	72.5	79.7	85.0	90.7	92.7	91.2	86.3	78.5	70.1	65.1	(23)
		MCWB	57.1	56.4	59.1	64.0	70.2	74.2	75.8	75.1	71.7	67.2	61.5	59.9	(24)
	10%	DB	58.7	61.5	67.9	76.0	82.0	88.0	90.0	88.6	83.3	75.1	66.3	60.7	(25)
		MCWB	52.2	53.5	56.8	62.6	68.7	73.1	75.4	74.4	70.7	66.0	59.2	54.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.0	65.2	66.8	71.0	75.9	78.4	80.1	79.9	77.3	75.6	69.1	67.2	(27)
		MCDB	68.8	71.3	75.2	80.6	85.5	90.1	92.1	91.7	86.0	81.5	72.0	71.7	(28)
	2%	WB	61.7	61.9	64.2	68.6	73.8	77.1	78.5	78.3	75.6	72.9	66.1	64.1	(29)
		MCDB	65.8	67.2	72.1	78.2	83.2	88.9	90.5	89.2	83.1	78.2	70.6	67.3	(30)
	5%	WB	58.1	58.6	62.0	66.6	72.1	76.0	77.6	77.2	74.3	70.2	63.5	60.6	(31)
		MCDB	62.4	64.4	69.3	75.7	81.2	86.9	89.0	87.2	81.0	76.0	68.3	64.7	(32)
	10%	WB	53.0	54.0	59.3	64.6	70.4	74.9	76.5	76.1	73.0	67.7	60.1	55.9	(33)
		MCDB	57.3	59.6	66.3	72.8	78.9	84.7	86.9	84.6	79.0	73.3	65.5	59.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	20.1	21.6	22.3	20.8	19.9	19.1	18.6	18.4	21.1	21.6	18.9	(35)
		MCDBR	23.6	25.2	26.0	25.9	23.5	23.1	22.5	21.9	22.0	23.8	25.1	22.4	(36)
	5% WB	MCWBR	17.0	16.6	13.9	12.2	10.2	8.2	7.2	7.0	8.2	12.0	15.2	16.0	(37)
		MCWBR	21.1	22.0	21.6	21.1	20.1	20.3	19.8	18.7	16.8	18.5	19.4	19.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.333	0.366	0.410	0.459	0.506	0.548	0.537	0.426	0.380	0.333	0.318	(40)
		taud	2.493	2.436	2.382	2.258	2.154	2.066	1.965	1.981	2.290	2.397	2.489	2.520	(41)
	Ebn at Noon		281	287	286	278	264	251	239	239	263	266	271	272	(42)
		Edh at Noon	26	31	36	43	48	53	58	56	39	32	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		812	1029	1357	1710	1893	2029	1914	1732	1442	1168	893	703	(44)
	RadStd		55	121	97	125	147	141	115	89	124	150	71	60	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+216	(46)
(47)	Regional (58 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+272	+184	(47)

Nomenclature: See separate page